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Red shirt color has no effect on winning in European Soccer: Reanalysis of Attrill et al. (2008) of the English premier league and six additional European leagues☆

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ABSTRACT

Attrill et al. (2008) conducted a far-reaching study in elite English soccer demonstrating in archival research that from 1946 to 2003 seasons, teams wearing red uniforms were more likely to win championships than teams in other uniform colors, won more at home and had a higher average league position (relative to cross-city rivals). Their study was one of only very few that extended the color-in-context theory (Elliot & maier, 2007) to team, ball-oriented long-duration sports. The current investigation returns to the red superiority hypothesis in professional soccer due to weaknesses in the original evidence for this hypothesis. We conducted two studies testing the red superiority hypothesis in professional soccer. In Study 1a, we first reanalyzed the original data and tested the strength of evidence in favor of the red superiority hypothesis. We then updated the English premier league data (1992–2018) and tested uniform color effects on game outcomes. In Study 2, we attempted to broaden the scope of Study 1 and increase statistical power by testing the red superiority effect during the last 20 years of six major European Soccer leagues (NOS Portugal, German Bundesliga, Dutch Eredivisie, Spanish la Liga, French Ligue 1, and the Italian Serie A). All three tests challenge the validity of the original findings and fail to detect uniform color effects at home play in professional soccer. In light of the current findings and a growing body of research in the field we call into question overall color effects in this athletic context.

Attrill et al. (2008) conducted commanding series of studies linking red uniform color to success in English soccer. Their study was archival, longitudinal and multi-faceted. Within the "Zeitgeist" of scrutinizing and reproducing scientific findings in the literature (e.g., Munafò et al., 2017; Nelson et al., 2018) the goal of the current investigation is to retest this important work and with the passage of time and the addition of data to conduct supplementary analyses.

Attrill et al. (2008) extended earlier research by Hupka, Zaleski, Otto, Reidl, and Tarabrina (1997) in Olympic combat sports, which showed red uniform superiority over blue in seemingly natural-experiment conditions. This early groundbreaking study was further supported by additional archival research (Falcó et al., 2016; Vasconcelos & Del Vecchio, 2017). Perception based experimental work of Elliot and Aarts (2011), Feltman and Elliot (2011) as well as Luck (2004) **lent** further credence to the red-superiority effect.

Elliot and Maier (2007) bemoaned the absence of theoretical foundation in psychological color effects and proposed their color-in-context (CIC) theory (Elliot & Maier, 2012). First, they argued that some colors carry a unique meaning, acquired through biological orientation reflected in evolutionary processes. Subsequently through cultural conditioning learned association might take place to reinforce the effect (for example, linking red with caution through repeated exposure to 'stop' signs for drivers or red error markings by teachers on tests). Lastly, Elliot and Maier theorized that the mere perception of color generated a fundamental evaluation of 'foe or friend' translated into a motivated behavior of either avoidance or approach, respectively. Similarly, in sport, it was proposed that encountering a red wearing rival should create tentativeness and hesitancy, which would lead to a competitive disadvantage.

Attrill et al. (2008) stands unique in support of the red superiority

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effect by expanding archival research into a sport environment that is fundamentally different than the original combat sport research (Hupka et al., 1997). For example, the duration of combat sports (e.g., free style wrestling, Greco-Roman wrestling) tends to be short while soccer is at least 90 min long. Combat sports by their very nature emphasize full-body brute force in addition to technique whereas, in soccer, sheer physical strength is of lesser sway on the outcome while lower-body ball control, movement and stamina are crucial. Finally, combat sports are individual based competitions whereas soccer involves the synchronization of 11 on-field players.

In Attrill et al. (2008), the researchers showed in an exhaustive set of archival studies that English soccer teams wearing red uniforms playing at their home venue during the 1946–2003 seasons were significantly more successful than other teams as well as more likely to win championships altogether. Commensurately, they did not find any differences in performance when the very same teams were playing at away venues when their uniform was different than their "traditional" red home uniform. Furthermore, they claimed that red teams were more successful against their non-red derby rivals (i.e., playing in the same large city) and in doing so argued that they had overcome the bias of "access to more resources in terms of fan base and financial resources than teams situated in smaller cities or towns" (p. 579).

The main goal of the present study was to return to the red superiority hypothesis in professional soccer that was introduced by the findings of Attrill et al. (2008) as a rapidly growing literature in psychology (especially in social psychology) has cast reasonable doubts on the robustness of initially published findings. More specifically, this literature has providing compelling evidence that many psychological findings simply do not hold up in direct and/or conceptual replication attempts and that prestigious journals seem to have favored innovation over rigor in the past (Camerer et al., 2018; Open Science Collaboration, 2015). Pertinent to the present research, a recent review on color effects in sports (Goldschmied et al., 2020) has demonstrated high levels of ambiguity and has highlighted several methodological shortcomings, including small sample sizes and questionable research practices (Asendorpf et al., 2013; Munafo et al., 2017; Nelson et al., 2018). Although, the field of sport psychology has begun to acknowledge the "so called replication crisis" in psychology (e.g. Schweizer & Furley, 2016), we are not aware of published replication attempts in this field. Hence, we conducted two studies testing the red superiority hypothesis in professional soccer. First, employing Attrill et al. (2008) as a foundation for the current analysis, in Study 1a, we first reanalyzed their data and reassessed their findings. We then applied their uniform color distribution and general statistical approach to the updated English premier league data (1992–2018; study 1b) and based on the original study predicted that red uniform wearing team would win more championships, be more successful at home play when they wear their traditional clothing and would be more successful when facing their geographically close rivals. In Study 2, we attempted to broaden the scope of Study 1 and increasing statistical power by testing the red superiority effect during the last 20 years of six major European Soccer leagues (NOS Portugal, German Bundesliga, Dutch Eredivisie, Spanish la Liga, French Ligue 1, and the Italian Serie A). We used the same approach as in Study 1 and Attrill et al. (2008) to test the central hypothesis that red uniform teams are more successful at home than teams wearing other colors.

1. Study 1a: Reanalysis of Attrill et al., 2008

1.1. Methods

As we did not receive the original data upon repeated requests by the original authors, we ran some further analyses based on Table 1 in Attrill et al. (2008). The distribution of jersey colors was: red (n = 16), blue (23), all white (11), and yellow-orange (9) and 'other' (9). As we could not follow the argument of the original authors of excluding one color category, we ran two separate analyses: One that included all five color categories and one identical to Attrill et al. (2008) that excluded the 'other' color category. Differences in performance of teams wearing each defined shirt color (red, blue, white, yellow-orange, 'other') were tested using a Kruskal-Wallis non-parametric test for differences between: % wins and Points/game as was conducted in the original study. We further ran Bayesian ANOVA to test the strength of evidence in favor of the null (no color effect on performance) or the alternative hypothesis (an effect of color on performance) with the same factors as above using the statistical software JASP (JASP Team, 2017). Bayes (BF10) factors were interpreted according to statistical convention (Lee & Wagenmakers, 2014; Raftery, 1995): Overall, BF_{10} ratios below the value of '1′ are in favor of the null hypothesis and above '1' in favor of the alternative hypothesis. However, BF_{10} ratios between 0.33 and 3 are typically treated as "weakly informative" (with 1.000 as not informative). Below 0.33, they are considered as moderate evidence in favor of the null hypothesis and below 0.1 as strong evidence for the null hypothesis. Furthermore, between 3 and 20 they are considered moderate evidence in favor of the alternative hypothesis, between 20 and 150 they are "strong" in favor of the alternative hypothesis.

1.2. Results

Winning Championships. Attrill et al. (2008) first argued that red wearing teams were more likely to win championships in English soccer. While they provided a remarkable figure, which showed red teams to be only 24% of all teams in contention, these teams won about 60% of all championship during the 1946–2003 seasons in seemingly strong support of the red superiority effect (all other uniform colors including "other" demonstrated the opposite trend with blue teams at 34% of all contenders, winning only about 20% the championships). However, the researchers did not conduct any statistical test to support this conclusion due to the data being non-independent from year to year.

We contend that the base line in the original study (i.e., labeled as "Total teams") was calculated erroneously. In essence each of the 68 teams in their sample, made an equal contribution to the "Total teams" color ratio (Red teams [24%] 16 out of 68, White teams [16%] 11 out of 68, Blue teams [34%], 23 out of 68, Yellow – Orange teams [13%] 9 out of 68 Other color teams [13%] 9 out of 68). As a case in point, Shrewsbury F.C. (seeded 67 in their ranking), a historically very marginal team made a comparable contribution as powerhouse clubs such as Liverpool or Manchester United. We maintain that the comparison level should be "weighted" by the number of seasons played for it to be considered as a valid base line.

Winning at Home. We conducted separate Kruskal-Wallis tests for the three English Leagues. For the English Premier League ($\chi2$ [4, N = 20] = 1.079, p = .898; with exclusion of 'other' category as in Attrill et al., 2008: $\chi2$ [3, N = 18] = 0.888, p = .828), and 3rd division ($\chi2$ [4, N = 24] = 3.572, p = .467; with exclusion of 'other' category as in Attrill et al., 2008: $\chi2$ [3, N = 21] = 3.590, p = .309) Kruskal-Wallis tests showed no significant differences between teams wearing the five shirt color categories in percent home wins in the Premier League and 3rd division. Only for the 2nd division ($\chi2$ [4, N = 24] = 10.099, p = .039, η^2 = 0.321 with exclusion of 'other' category as in Attrill et al., 2008: $\chi2$ [3, N = 20] = 9.104, p = .028, η^2 = 0.381) a Kruskal-Wallis test revealed a significant effect. However, follow-up Bonferroni-corrected pairwise comparisons only showed a restricted significant difference between the red and the blue category (p = .035).

Bayesian ANOVA. Overall the former analyses provided very limited evidence for the alternative hypothesis of an effect of jersey color on the percentages of home wins. A Bayesian ANOVA on the percentages of home wins in the premiere league (BF_{10} = 0.351; 0.397 with exclusion of the 'other' category), second division (BF_{10} = 1.000; 1.000 with exclusion of the 'other' category), and third division (BF_{10} = 0.353; 0.521 with exclusion of the 'other' category) revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins

Table 1
EPL Team ranking (1992–2018) based on percentage wins at home and points per game.

Ranking	Team Name	Ranking in Attrill et al. (2008)	Uniform Color	% wins	Points Per Game
1	Man Utd (27)	2	R	71.89	2.34
2	Arsenal (26)	4	R	65.33	2.18
3	Chelsea (27)	62	B	62.85	2.12
4	Liverpool (27)	1	R	61.67	2.1
5	Man City (22)	23	B	56.22	1.9
6	Tottenham (27)	11	W	53.34	1.84
7	**Newcastle (24)**	7	**O (B)**	**50.92**	**1.77**
8	Leeds (12)	3	W	50.17	1.77
9	Blackburn (18)	10	B	48.19	1.7
10	Everton (27)	21	B	47.14	1.68
11	Fulham (14)	68	W	44.36	1.55
12	West Ham (23)	46	O (M)	43.38	1.55
13	Stoke (10)	44	R	43.27	1.57
14	Southampton (20)	8	R	40.9	1.52
15	Aston Villa (24)	17	O (M)	40.88	1.52
16	Portsmouth (7)	NI	B	40.6	1.47
17	Reading (3)	5	B	40.35	1.42

18	Sheffield Weds (8)	47	B	39.49	1.5
19	Wimbledon (8)	NI	B	38.91	1.47
20	Birmingham (8)	66	B	38.6	1.49
21	Swansea (7)	29	W	38.35	1.43
22	Middlesborough (15)	20	R	38.33	1.45
23	Leicester (13)	NI	B	38.27	1.45
24	Bournemouth (4)	19	R	38.16	1.39
25	Charlton (8)	NI	R	38.16	1.41
26	Bolton (14)	28	W	38.16	0.3
27	Sheffield Utd (3)	17	R	38.1	1.52
28	Norwich (8)	56	Y	37.25	1.45
29	Barnsley (1)	52	R	36.84	1.32
30	Burnley (6)	25	O (M)	36.84	1.34
31	Coventry (9)	59	B	36.73	1.42
32	QPR (7)	13	B	36.2	1.37
33	Oldham (3)	58	B	35.72	1.4
34	Notts Forest (5)	36	R	35.04	1.38
35	Ipswich (5)	6	B	35.04	1.32
36	Wolves (5)	12	Y	34.74	1.27
37	Brighton (2)	25	B	34.21	1.37
38	Watford (6)	61	Y	32.46	1.25
39	W. Bromwich (11)	NI	B	32.06	1.22
40	Sunderland (16)	45	R	31.93	1.25
41	Wigan (8)	NI	B	31.58	1.24
42	Cardiff (1)*	62	B	31.58	1.05
43	Hull (3)	60	Y	30.53	1.17
44	Derby (7)	27	W	30.09	0.26
45	Crystal Palace (4)	NI	O (R/B)	29.57	1.13
46	Bradford City (2)	49	Y	26.32	1.18
47	Blackpool (1)	64	Y	26.32	1.05
48	Huddersfield (2)	55	B	21.06	0.84
49	Swindon (1)	30	R	19.05	0.9

and weak support for the null hypothesis of no color effect.

Combined analyses collapsed over the three English leagues. For all 68 teams collapsed over the three English leagues a Kruskal-Wallis tests on the percent home games won ($\chi 2$ [4, N = 68] = 9.025, p = .060; with exclusion of 'other' category as in Attrill et al., 2008: $\chi 2$ [3, N = 59] = 8.406, p = .038, η^2 = 0.098) showed no significant differences between teams wearing the five shirt color categories. A Bayesian ANOVA on the percentages of home wins for all three English leagues (BF_{10} = 1.000; 1.000 with exclusion of the 'other' category) revealed neither evidence for the alternative hypothesis nor the null hypothesis.

Points per game at Home. We conducted separate Kruskal-Wallis tests for the three English Leagues. For the English Premier League ($\chi 2$ [4, N = 20] = 1.973, p = .741; with exclusion of 'other' category as in Attrill et al., 2008: $\chi 2$ [3, N = 18] = 1.268, p = .737), 2nd ($\chi 2$ [4, N = 24] = 9.446, p = .051; with exclusion of 'other' category as in Attrill et al., 2008: $\chi 2$ [3, N = 20] = 8.771, p = .032, η^2 = 0.361), and 3rd division ($\chi 2$ [4, N = 24] = 3.528, p = .474; with exclusion of 'other' category as in Attrill et al., 2008: $\chi 2$ [3, N = 21] = 3.527, df = 3, p = .317) Kruskal-Wallis tests showed no significant differences between teams wearing the five shirt color categories in points per home game (except in the second division when excluding the 'other' color category).

Bayesian ANOVA. A Bayesian ANOVA on the average points per home game in the premiere league (BF_{10} = 0.401; 0.434 with exclusion of the 'other' category), second division (BF_{10} = 1.000; 1.000 with exclusion of the 'other' category), and third division (BF_{10} = 0.458; 0.627 with exclusion of the 'other' category) revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins and weak support for the null hypothesis of no color effect.

Combined analyses collapsed over the three English leagues. For all 68 teams collapsed over the three English leagues a Kruskal-Wallis tests on the average points per home game ($\chi 2$ [4, N = 68] = 9.934, p = .042, η^2 = 0.094; with exclusion of 'other' category as in Attrill et al., 2008: $\chi 2$ [3, N = 59] = 8.859, p = .031, η^2 = 0.090) showed a significant difference between teams wearing the five shirt color categories. However, follow-up Bonferroni-corrected pairwise comparisons only showed a significant difference between the yellow/orange and the red category (p = .024).

A Bayesian ANOVA on the average points per home game for all three English leagues (BF_{10} = 1.000; 1.000 with exclusion of the 'other' category) revealed neither evidence for the alternative hypothesis nor the null hypothesis.

1.3. Discussion

Our reanalyzes of Attrill et al. (2008) provide very limited support for the conclusions drawn by the original authors. We also had some further concerns regarding the sampling choices in the original study that raise concerns regarding questionable research practices and p-hacking (Munafo et al., 2017; Nelson et al., 2018). We therefore attempted to collect additional data and this time only include data from the start of the English Premiere League until the 2018/19 (end of data collection for this study). The founding of the EPL in 1992 was an inflection point in English soccer whereupon it transitioned from regional-parochial endeavor to a global enterprise (Robinson & Clegg, 2018). We utilized the establishment of the league as the genesis for the current analysis as relative uniformity in game rules and governance has prevailed since. While the playing field is certainly not level for the clubs participating in the EPL there are several mechanisms in place, which did not exist before its establishment, to sustain some level of parity between the clubs. For example, all teams earn a flat participation fee and flat commercial rights fee (The league sells its television rights on a collective basis, half of the sum is distributed evenly between the teams). In addition, there had been some substantial changes in rules and lighting conditions during the period analyzed in Attrill et al. (2008) that arguably might affect color effects on football performance.

Rule Changes. The proposal to increase the points awarded to winners from 2 to 3 was approved in 1981 by the FA. The change in points awarded to the winner from the subsequent season had an effect on the proportion of wins (W), home team wins (HW), visiting team wins (VW) and ties (T) as well as relative wins and ties percentages (Palacios-Huerta, 2004). For example, the home team wins (HW) for the 1947–1982 period was .503 but declined precipitously to .474 during the 1983–1996 period. This rule change according to Palacios-Huerta's analysis reflected a structural change similar to the effects that the stoppage of play brought about by WWII (Home team wins (HW) for the 1920–1939: 0.547). This fundamental change in play is not reflected in Attrill et al. (2008) as all data standardized as 3 points for a win to all games regardless of season play.

Light conditions during matches. From 1930 to 1950 the FA (Football Association) banned floodlit soccer matches but then reversed course. In the 1956 season, Portsmouth played Newcastle in the first official game with floodlights. With growing TV broadcasting, this technology aiding in players' visibility and performances as well as their supporters' enjoyment became paramount and within several seasons, all first league stadiums had floodlights (Cox et al., 2002). The introduction of artificial illumination may have altered visibility and color perception in rain or clouds conditions. Olde Rikkert et al. (2015) demonstrated experimentally that the color of a soccer outfit affected visibility. It is also important to note that research in soccer is unique in being conducted outdoors as almost all past archival studies to support uniform color effects were explored in indoor settings (Falcó et al., 2016; Hupka et al., 1997; Vasconcelos & Del Vecchio, 2017).

2. Study 1b: red superiority in the English Premier League?

2.1. Methods

We replicated Attrill et al. (2008) in our data gathering method, classification approach and analysis to the extent possible. When we deviated from their original study, we made sure to highlight the different approach undertaken.

We utilized the extensive on-line data available through the EPL archives (https://www.premierleague.com/) to investigate the relative success of EPL teams since the 1992-93 season when the league was established until the 2018-19 season, the last season played in full under non-pandemic conditions. Average ranking of geographically close rivals as well as derby winning analysis were the exceptions as we were able to combine records since 1946.

To determine whether red teams had been more successful within the EPL since its inception, we investigated the winning performance of each team in matches played at their home venue, expressed as percentage wins over all games played at home [% wins] and mean points per home game ('3' for a win and '1' for a draw) [Points/game]. All the teams that played at least one season of play were included in the analysis (the number of seasons played by each club is included in Table 1 with their % wins and Points/game).

Clubs wear their signature colors in home games, but change to an alternative in away games when there is a color clash. Hence, the colors worn during away games are not consistent.

The dominant home shirt color for these 49 clubs over the time period in the EPL was determined and categorized (using http://www.historicalkits.co.uk/index.htm). The final distribution for the overall sample of teams was: red (n = 14), blue (18), all white (6), and yellow-orange (6), and other (4). Where a shirt was mainly one color plus some white, the team was classified under the color. Yet, there are two exceptions of note in this classification system: Crystal Palace, was categorized as "other" since it had an about even mix of red and blue and Bradford F.C. was classified as yellow-orange though it has a striped uniform of yellow and claret (we did so in line with the original research). As in the previous analyses we included both tests with four color categories as in Attrill et al. (2008) and tests with all five categories.

Differences in performance of teams wearing each defined shirt color (red, blue, white, yellow-orange, 'other') were then tested using a Kruskal-Wallis non-parametric test for differences between: % wins and Points/game. We again followed these tests with Bayesian ANOVAs. Lastly, we conducted a series of Wilcox singed rank tests for historical derbies in English soccer whereupon one team was red wearing.

2.2. Results

Winning championships. For the current study, (1992–2018), only red and blue wearing teams won the EPL championship so we consider only these two colors in our championship analysis. While the 14 red wearing teams to compete during this time-span had the combined 165 seasons of play (47%), the 18 blue wearing teams had 188 seasons of play (53%). During this period, red teams won 16 championships (59%) while blue ones won 11 (41%). Indeed, while the data cannot be subjected to a reliable statistical analysis due to seasonal non-independence, the success disparity based on uniform color is much reduced in comparison to the original study. In summary, neither Attrill et al. (2008) nor the current analysis could have provided credence to the red superiority effect.

Winning at Home. We conducted a Kruskal-Wallis test for our Premier League data (1992–2018 seasons). A Kruskal-Wallis tests with the exclusion of the 'other' category as in Study 1 and Attrill et al., 2008 on the percent home games won ($\chi2$ [4, $N = 49$] = 7.822, $p = .098$; $\chi2$ [3, $N = 44$] = 7.596, = 3, $p = .055$), the average points at home ($\chi2$ [4, $N = 49$] = 4.840, $p = .304$; $\chi2$ [3, $N = 44$] = 4.649, $p = .199$) [see Fig. 1], and the average ranking ($\chi2$ [4, $N = 49$] = 8.871, $p = .064$; $\chi2$ [3, $N = 44$] = 7.998, $p = .046$, $\eta^2 = 0.125$) showed no significant differences between teams wearing the five shirt color categories (only for average rank

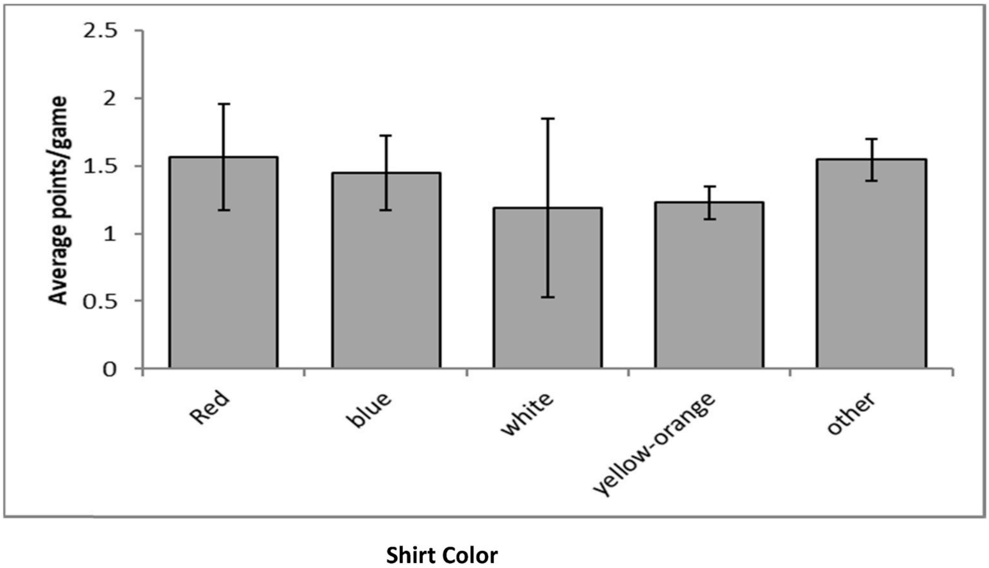

Fig. 1. Mean number of points per home game achieved by teams of each shirt color of all EPL teams wearing each shirt color at home from 1992 to 2018. Other shirt color not included in the analysis.

when excluding the 'other' category). However, there was only a significant difference between white playing teams and yellow/orange teams with Bonferroni-corrected pairwise comparisons, indicating that yellow/orange teams ranked worse than white teams).[2] **In order to increase the power of our test, we also conducted a Mann-Whitney test contrasting teams in red uniforms vs. the rest of the teams (including the 'other' category). Teams wearing red (M = 23.54) did not rank higher than all others (M = 25.59), [Z = -0.45, p = .65, eta squared = 0.004], nor did red score more points per home games than all other teams, [Z = -0.69, p = .49, eta squared = 0.001].**

Bayesian ANOVA. A Bayesian ANOVA on the average wins, (BF_{10} = 0.328; 0.452 with the exclusion of the 'other' category), points per home game (BF_{10} = 0.368; 0.499 with the exclusion of the 'other' category), average ranking (BF_{10} = 0.420; 0.549 with the exclusion of the 'other' category) for the new Premier League data revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins, points per game, and average ranking and weak to moderate support for the null hypothesis of no color effect.

Average Ranking of Geographically Close Rivals. We further reassessed the original study's derby comparison analysis. The researchers noted that it was conducted to exclude the "wealth" bias, as the most successful red soccer teams were based in large metropolitan areas and thus may have access to more resources than teams located in smaller cities or towns. Their approach seemed plausible but was lacking in two major aspects: sampling bias and a questionable dependent variable. For the first, Attrill et al. (2008) practiced convenience sampling which is not intended for to create generalizations pertaining to the entire population (Etikan et al., 2016). It is not clear how these particular derby comparisons were selected and while some are unquestionably of great popularity and prominence (e.g., Liverpool vs. Everton), others may not be (e.g., Charlton vs. Millwall in South-East London). Other derbies of tradition and distinction were altogether excluded (e.g., Sunderland vs. Newcastle United). The second problem of the derby comparison analysis pertained to the outcome variable, average league position differential (red team – other team). This dependent variable is problematic since it does not take into account the strength of each team relative to its geographical rival, which should be measured by games played against each other. Instead this outcome variable measures the success of the team relative to all its competitors in the division. Thus, absurdly, a team playing in the second division and ranked at the top of its league beating other teams with similar financial resources is ranked lower than a team competing at the bottom of the top league and failing against teams with comparable economic means.

Since many of the derby teams did not play in the EPL nor did they play in the same league in every season, we added seasons 2004–2018 to the original data (1946–2003) and reserved the analysis only to seasons when the teams played in the same league. We also added the Tyne–-Wear derby (inter-city rivalry in North East England with the two cities of Sunderland [red unfirms] and Newcastle United [black unfirms] just 12 miles apart as the British popular press found this rivalry to be one of the most prominent in English soccer (Zed, 2017). These two teams were also included in Attrill et al. (2008) best original 68 teams. We conducted a series of Wilcox singed rank tests for each derby.

Among the comparisons which supported the red superiority effect, Liverpool (red, M = 3.93, SD = 2.73) had a better average ranking then Everton (blue, M = 9.23, SD = 5.37) in 61 seasons (Z = −5.93, p = .00, eta squared = 0.59), Manchester United (red, M = 4.98, SD = 4.74) had a better average ranking then Manchester City (blue, M = 10.66, SD = 6.48) in 61 seasons (Z = −4.60, p = .00, eta squared = 0.35), in the North London derby, Arsenal (red, M = 5.79, SD = 3.98) did have a better average ranking than Tottenham (white, M = 7.93, SD = 4.92) in 68 seasons (Z = −2.47, p = .014, eta squared = 0.09), Nottingham Forest (red, M = 7.33, SD = 5.55) did have a better average ranking than Notts County (black, M = 14.06, SD = 6.45) in 18 seasons (Z = −2.68, p =

[2] We also conducted a Kruskal-Wallis test for the combined data of Attrill et al.'s Premier League teams and the new data starting after the Attrill et al. data collection in 2003 until 2018/19). A Kruskal-Wallis tests on the percent home games won ($\chi2$ [4, N = 47] = 6.130, p = .190; $\chi2$ [3, N = 42] = 5.004, p = .172 with the exclusion of the 'other' category as in Study 1 and Attrill et al., 2008), the average points at home ($\chi2$ [4, N = 47] = 6.531, p = .163; $\chi2$ [3, N = 42] = 5.260, p = .154 with the exclusion of the 'other' category), and the average ranking ($\chi2$ [4, N = 47] = 4.558, p = .336; $\chi2$ [3, N = 42] = 4.221, p = .239 with the exclusion of the 'other' category) showed no significant differences between teams wearing the five shirt color categories.A Bayesian ANOVA on the average wins, (BF10 = 0.406; 0.467 with the exclusion of the 'other' category), points per home game (BF10 = 0.426; 0.469 with the exclusion of the 'other' category), average ranking (BF10 = 0.281; 0.365 with the exclusion of the 'other' category) for the combined Premier League data revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins, points per game, and average ranking and weak to moderate support for the null hypothesis of no color effect.

.007, eta squared = 0.42). However, five comparisons of the nine undertaken, failed to show the red superiority effect. Bristol City (red, M = 9.97, SD = 5.98) did not have a better average ranking than Bristol Rovers (blue, M = 11.21, SD = 6.40) in 33 seasons (Z = −0.68, p = .5, eta squared = 0.01), In the South-East London Derby, Charlton (red, M = 13.15, SD = 5.5) did not have a better average ranking than Millwall (blue, M = 11.19, SD = 6.28) in 26 seasons (Z = −1.46, p = .14, eta squared = 0.08), Sheffield United (red, M = 9.45, SD = 5.58) did not have a better average ranking than Sheffield Wednesday (blue, M = 11.88, SD = 6.79) in 33 seasons (Z = −1.51, p = .13, eta squared = 0.07), Stoke (red, M = 10.3, SD = 7.96) did not have a better average ranking than Port Vale (blue, M = 12.4, SD = 4.72) in 10 seasons (Z = −0.97, p = .33, eta squared = 0.1) and Sunderland (red, M = 12.24, SD = 6.79) did not have a better average ranking than New Castle (black, M = 9.73, SD = 5.45) in 45 seasons (Z = −1.82, p = .069, eta squared = 0.07). All in all, while the omnibus test was significant only a minority of the comparisons (now only four of the nine) were showing the red superiority effect (see Fig. 2).

Winning Derbies. In order to address the convoluted nature of average ranking as the outcome variable, we conducted a more straightforward analysis of derby match outcome since 1946 until the 2018/9 season not including friendly matches and whenever the game was not played at the home venue of one of the teams. We kept with the original Attrill study (2008) of awarding 3 points for a win by the home team throughout. A series of Mann-Whitney tests showed that Liverpool (red, M = 1.9, SD = 1.16) performed better at home than Everton (blue, M = 1.11, SD = 1.19) in 148 games (Z = −4.03, p = .00, eta squared = 0.11). In the North London derby, Arsenal (red, M = 1.86, SD = 1.25) gained more points at home than Tottenham (white, M = 1.45, SD = 1.31) in 157 games (Z = −2.01, p = .045, eta squared = 0.03). However, Manchester United (red, M = 1.73, SD = 1.29) did not perform better at home than Manchester City (blue, M = 1.39, SD = 1.32) in 130 derbies (Z = −1.55, p = .12, eta squared = 0.02). Bristol City (red, M = 1.77, SD = 1.27) did not gain more points at home than Bristol Rovers (blue, M = 1.37, SD = 1.33) in 133 matches (Z = −1.83, p = .07, eta squared = 0.02). Sunderland (red, M = 1.38, SD = 1.17) did not do better than New Castle (black, M = 1.57, SD = 1.25) in 84 games (Z = −0.61, p = .54, eta squared = 0.004). Sheffield United (red, M = 1.7, SD = 1.21) did not surpass Sheffield Wednesday (blue, M = 1.64, SD = 1.3) in 61 derbies (Z = −0.26, p = .79, eta squared = 0.001). Nottingham Forest (red, M = 1.22, SD = 1.35) did not gain more points at home than Notts County (black, M = 1.29, SD = 1.21) in 35 derby matches (Z = −0.42, p = .71, eta squared = 0.01). Stoke (red, M = 1.31, SD = 1.25) did not perform better than Port Vale (blue, M = 1.46, SD = 1.13) in 29 games (Z = −0.52, p = .65, eta squared = 0.01). Lastly, in reversal in the South-East London Derby, Millwall (blue, M = 1.88, SD = 1.18) outperformed Charlton (red, M = 0.96, SD = 1.04) in 50 derbies (Z = −2.75, p = .006, eta squared = 0.15). All in all, while the omnibus test was still significant only two of the comparisons were showing the red superiority effect, six in-city rivalries failed to do so and one was showing a reversal and the overall effect size was small (Z = −3.06, p = .002, eta squared = 0.01).

2.3. Discussion

The results of Study 1a and 1b showed that the Attrill et al. (2008) were flawed and hardly warranted the authors' conclusions. First, analysis could not have been performed in historical championship inquiry due to non-independence. Second, post-hoc comparisons were not conducted for the main performance variables. Third, the derby analysis did suffer from convenience sampling methodology as well as a compromised outcome variable. Still, if one is to accept the red superiority effect finding despite these fundamental failings, the association is likely to be the product of a spurious association due to the limited sample size (n = 68), as three of the most prominent teams in contention (i.e., Liverpool, Manchester United, and Arsenal) just happen to wear red uniforms. The impact of these teams weighted even more extremely in the average ranking comparisons, as they comprised three out of the eight original comparisons (almost half of the comparisons showing the effect).

Another problem with the original Attrill et al. (2008) pertains to their color classification. The researchers generated an "other" category and lumped together teams that were not wearing red, blue, white or yellow/orange. This category included also uniforms, which were a combination of colors without a dominant one. While this classification made sense in light of the low numbers of teams wearing green (Plymouth) or black (Newcastle United and Notts County) and was excluded from their main analysis, it is problematic in light of the psychological meaning and the potential advantage these colors bestow in athletic competition based on past research. For example, Sorokowski et al (2014) found a black superiority effect. The color green was also found to have a unique psychological influence (Lichtenfeld et al., 2012). Also of concern is prominent teams wearing claret uniforms (e.g., Aston Villa, Burnley, West Ham) being lumped into the "other" category while being a shade of red (Horiguchi & Iwamatsu, 2018). When we included this category regardless, we still failed to detect the effect.

For the most updated and comparable data analyzed in Study 1b

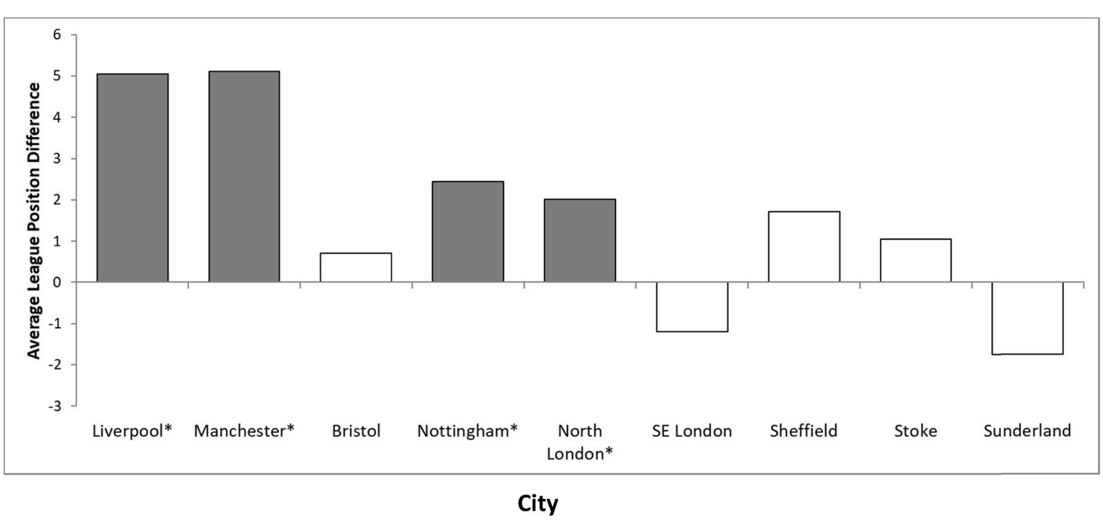

Fig. 2. The difference in performance between teams from the same city where one of these teams wears red shirts as their home color. The results are presented as the number of league places separating the two teams playing in the same league when mean league position since 1947 is calculated. This is expressed as the red team average position minus the average position of the other team within the city. Therefore, positive values show a higher average position for the red team.

(1992–2018) we find that red uniform did not confer advantage in the EPL. All uniform colors performed similarly in the number of points gained at home as well as % wins. Due to the aforementioned sample size limitation coinciding with low statistical power in Study 1 **as well as some other limitations plaguing the current research akin to the original study (i.e., historical analysis of derby play without accounting for rule changes or lighting conditions)**, we attempted to conduct a higher-powered test of red superiority in Study 2 by analyzing **recent** performance effects as a function of uniform color in six major European leagues.

3. Study 2: red superiority in European Soccer?

3.1. Methods

The data gathering method, classification approach and analysis was analogous to Study 1. We planned to sample at least 200 teams for the main analyses, which would provide sufficient power (0.80) to obtain medium effects ($f = 0.2$) in the 1×5 (color categories) between-subject design (Faul et al., 2007; cf.; Schweizer & Furley, 2016). We had to include the additional category ‘other color’ compared to the four categories ‘red’ (28.1% of teams, total 65 teams), ‘blue’ (26% of teams, total 60 teams), ‘white’ (17.3% of teams, total 40 teams), and ‘yellow/orange’ (11.7% of teams, total 27 teams) as 39 teams wore a different jersey color (16.9%; 16 teams wore green, 17 black, 1 copper, 3 violet, and 2 pink) and did not fit the categories of Attrill et al. (2008). Hence, we had five color categories as opposed to the four categories of Atttril et al. and Study 1.[3] We utilized the extensive on-line data available (https://www.soccerstand.com/) to investigate the relative success of the Teams in the six major European Soccer leagues (NOS Portugal, German Bundesliga, Dutch Eredivisie, Spanish la Liga, French Ligue 1, and the Italian Serie A) in the last 20 years since the 1999–2019 seasons. Our final sample was 231 teams (37 Portuguese teams, 37 German teams, 30 Dutch teams, 42 Spanish teams, 40 French teams, and 45 Italian teams). If the relevant data could not be obtained from this primary source of information we used additional sources such as the club homepages or additional webpages (e.g., youtube.com, oldfootballshirts.com, footballkitarchive.com, etc.). To assure accuracy of data every data point found on a website was verified using a secondary source. All the teams that played at least one season of play were included in the analysis (the number of seasons played by each club is included in the online supplements with their % wins, average home points, average home wins, and average ranking). Differences in performance of teams wearing each defined shirt color (red, blue, white, yellow-orange, other) were tested using a Kruskal-Wallis non-parametric test for differences between: % wins, Points/game, average ranking. We again followed up these tests with Bayesian ANOVAs.

3.2. Results

Winning at Home. We conducted separate Kruskal-Wallis tests for the six European Leagues (see Fig. 3 for a summary of the descriptive statistics). For the Portuguese ($\chi2$ [4, N = 37] = 2.004, p = .735), German ($\chi2$ [4, N = 37] = 3.122, p = .538), Dutch ($\chi2$ [4, N = 30] = 4.589, df = 4, p = .332), Spanish ($\chi2$ [4, N = 42] = 1.264, p = .868), and French leagues ($\chi2$ [4, N = 40] = 2.545, p = .637) Kruskal-Wallis tests showed no significant differences between teams wearing the five shirt color categories in percent home wins. Only for the Italian league ($\chi2$ [4, N = 45] = 9.964, p = .041, η^2 = 0.149) a Kruskal-Wallis test revealed a significant effect. However, follow-up pairwise comparisons only showed significant differences between the yellow/orange and the blue category (p = .026; Bonferroni-corrected: p = .256) and the white and blue category (p = .011; Bonferroni-corrected: p = .114) and none of the comparisons remained significant after a Bonferroni correction for multiple tests.

Bayesian ANOVA. A Bayesian ANOVA on the average home game wins in the Portuguese (BF_{10} = 0.164; 0.244 with the exclusion of the ‘other’ category), German (BF_{10} = 0.533; 0.400 with the exclusion of the ‘other’ category), Dutch (BF_{10} = 0.867; 0.861 with the exclusion of the ‘other’ category), Spanish (BF_{10} = 0.171; 0.188 with the exclusion of the ‘other’ category), French (BF_{10} = 0.235; 0.252 with the exclusion of the ‘other’ category), and Italian league (BF_{10} = 1.000; 1.000 with the exclusion of the ‘other’ category) revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins and weak to moderate support across the six leagues for the null hypothesis of no color effect.

Points per game at Home. We conducted separate Kruskal-Wallis tests for the six European Leagues on the average points per home game (see Fig. 4 for a summary of the descriptive statistics). For the Portuguese ($\chi2$ [4, N = 37] = 2.294, p = .682), German ($\chi2$ [4, N = 37] = 3.624, p = .459), Dutch ($\chi2$ [4, N = 30] = 4.757, p = .313), Spanish ($\chi2$ [4, N = 42] = 1.456, p = .834), and French leagues ($\chi2$ [4, N = 40] = 2.176, p = .703) Kruskal-Wallis tests showed no significant differences between teams wearing the five shirt color categories in percent home wins. Only for the Italian league ($\chi2$ [4, N = 45] = 9.571, p = .048, η^2 = 0.139) a Kruskal-Wallis test revealed a significant effect. However, follow-up pairwise comparisons only showed significant differences between the yellow/orange and the blue category (p = .035; Bonferroni-corrected: p = .349) and the white and blue category (p = .010; Bonferroni-corrected: p = .100) and none of the comparisons remained significant after a Bonferroni correction for multiple tests.

Bayesian ANOVA. A Bayesian ANOVA on the average points per home game in the Portuguese (BF_{10} = 0.161; 0.242 with the exclusion of the ‘other’ category), German (BF_{10} = 0.627; 0.398 with the exclusion of the ‘other’ category), Dutch (BF_{10} = 0.867; 0.879 with the exclusion of the ‘other’ category), Spanish (BF_{10} = 0.172; 0.190 with the exclusion of the ‘other’ category), French (BF_{10} = 0.235; 0.294 with the exclusion of the ‘other’ category), and Italian league (BF_{10} = 1.000; 1.000 with the exclusion of the ‘other’ category) revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins and weak to moderate support across the six leagues for the null hypothesis of no color effect.

Average rank per season. We conducted separate Kruskal-Wallis tests for the six European Leagues on the average ranking of the teams (see Fig. 5 for a summary of the descriptive statistics). For the Portuguese ($\chi2$ [4, N = 231] = 3.003, p = .557), German ($\chi2$ [4, N = 37] = 3.182, p = .528), Dutch ($\chi2$ [4, N = 30] = 4.678, p = .322), Spanish ($\chi2$ [4, N = 42] = 0.911, p = .923), and French leagues ($\chi2$ [4, N = 40] = 1.899, p = .754) Kruskal-Wallis tests showed no significant differences between teams wearing the five shirt color categories in percent home wins. Only for the Italian league ($\chi2$ [4, N = 45] = 9.579, p = .048, η^2 = 0.139) a Kruskal-Wallis test revealed a significant effect. However, follow-up pairwise comparisons only showed significant differences between the other and the white category (p = .013; Bonferroni-corrected: p = .133) and the white and blue category (p = .018; Bonferroni-corrected: p = .177) and none of the comparisons remained significant after a Bonferroni correction for multiple tests.

Bayesian ANOVA. A Bayesian ANOVA on the average ranking in the Portuguese (BF_{10} = 0.193; 0.256 with the exclusion of the ‘other’ category), German (BF_{10} = 0.267; 0.260 with the exclusion of the ‘other’ category), Dutch (BF_{10} = 0.867; 0.918 with the exclusion of the ‘other’ category), Spanish (BF_{10} = 0.162; 0.177 with the exclusion of the ‘other’ category), French (BF_{10} = 0.194; 0.253 with the exclusion of the ‘other’ category), and Italian league (BF_{10} = 1.000; 1.000 with the exclusion of the ‘other’ category) revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins and weak to moderate

[3] We run the same analyses in a 1×4 and a 1×5 (color categories) between-subject design to check if including the ‘other-category’ as opposed to Attrill et al (2008) and Study 1 would influence the pattern of results. This was not the case as no additional significant effects emerged.

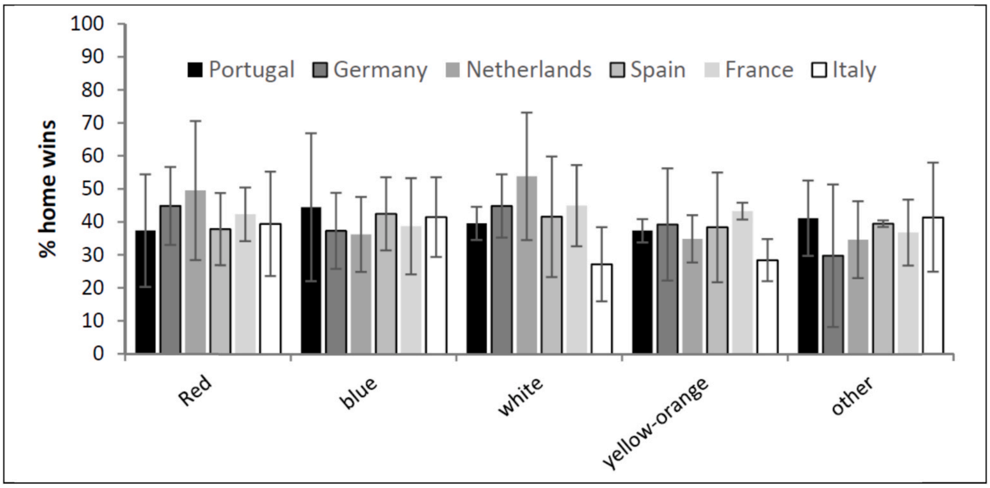

Fig. 3. Percentage of home game wins achieved by teams of each shirt color of the respective European teams wearing each shirt color at home from 1999 to 2019 seasons. Error bars represent Standard Deviations.

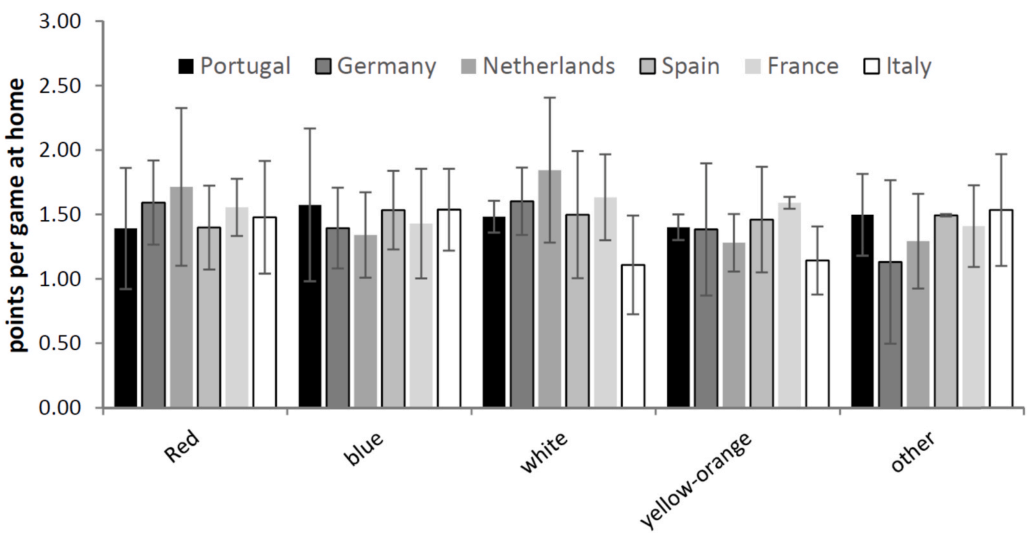

Fig. 4. Mean number of points per home game achieved by teams of each shirt color of the respective European teams wearing each shirt color at home from 1999 to 2019 seasons. Error bars represent Standard Deviations.

support across the six leagues for the null hypothesis of no color effect.

Combined analyses collapsed over the six European leagues. For all 231 teams collapsed over the six European leagues a Kruskal-Wallis tests on the percent home games won ($\chi2$ [4, $N = 231$] $= 4.405$, $p =$ .354; $\chi2$ [3, $N = 192$] $= 3.998$, $p = .262$ with the exclusion of the 'other' category as in Study 1 and Attrill et al., 2008), the average points at home ($\chi2$ [4, $N = 231$] $= 4.611$, $p = .330$; $\chi2$ [3, $N = 192$] $= 4.286$, $p =$ .232 with the exclusion of the 'other' category), and the average ranking ($\chi2$ [4, $N = 231$] $= 3.010$, $p = .556$; $\chi2$ [3, $N = 192$] $= 2.254$, $p = .521$ with the exclusion of the 'other' category) showed no significant differences between teams wearing the five shirt color categories.

Bayesian ANOVA. A Bayesian ANOVA on the average wins, ($BF_{10} =$ 0.076; 0.118 with the exclusion of the 'other' category), points per home game ($BF_{10} = 0.088$; 0.145 with the exclusion of the 'other' category), average ranking ($BF_{10} = 0.046$; 0.097 with the exclusion of the 'other' category) across all European Leagues revealed no evidence for the alternative hypothesis of a color effect on percentages of home wins, points per game, and average ranking and strong support for the null hypothesis of no color effect.

3.3. Discussion

The results of Study 2 show no indication that wearing the color red in professional soccer might be associated with performance advantages. Across the six most prestigious European Leagues uniform colors did not significantly impact soccer performance in the last 20 years. Bayesian analyses provided strong support for the null hypothesis suggesting that wearing red (or any other color) was not associated with any measurable performance advantage in professional European soccer.

4. General discussion

The goal of the present research was initially to retest Attrill et al. (2008) in line with the tradition in sport research to reevaluate the validity of past findings with more recent data (Baumeister & Steinhilber, 1984; Morgulev & Galily, 2018; Schlenker et al., 1995) and a larger, more comprehensive sample (Schweizer & Furley, 2016). As increasing research within psychological science in general (e.g. Munafo et al., 2017; Open Science Collaboration, 2015) and research on the red superiority effect in sports in particular (Goldschmied et al., 2020; Goldschmied & Lucena, 2018; Goldschmied & Spitznagel, 2020) has cast

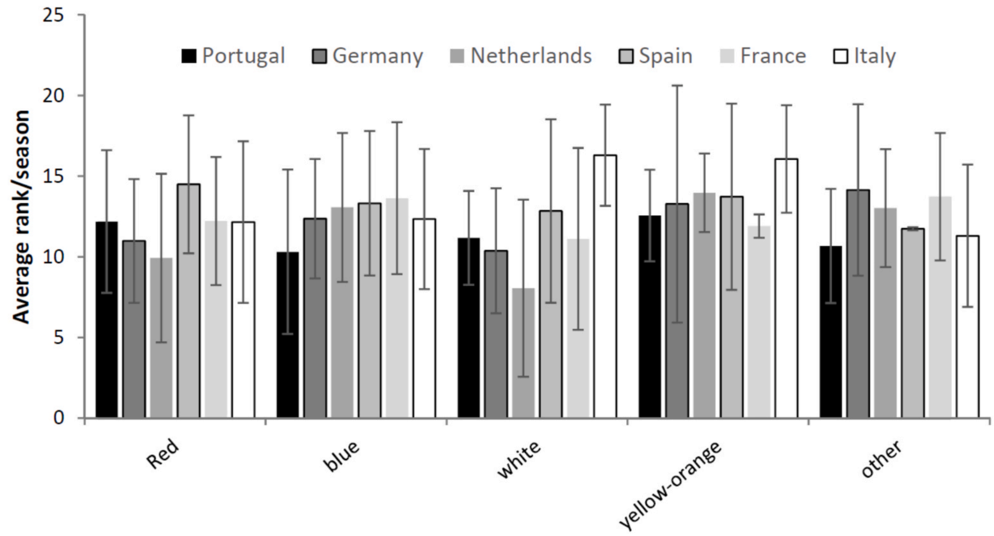

Fig. 5. Mean league rank achieved by teams of each shirt color of the respective European teams wearing each shirt color at home from 1999 to 2019 seasons. Error bars represent Standard Deviations.

doubt on the reliability of a substantial number of published reports. We consider this research to be not only important but necessary. The present analyses provide no indication that wearing a certain uniform color is consistently associated with superior or inferior performance in English soccer in particular and elite European soccer overall.

Past archival research, exploring uniform color effects in team sports is mostly in-line with the results of the present research. For example, Piatti et al. (2012) conducted research in Australian rugby, another outdoors and ball-oriented sport (albeit with a significant aggression element) during a period extending 30 seasons ($N = 5604$ games) for the 36 teams competing in the country's elite league. They detected the red superiority effect in a direct binomial contrast but when applying advanced multivariate statistics, the effect was no longer apparent suggesting that it may have been propelled by a spurious association (i. e., very successful teams wearing red by happenstance). The researchers themselves went on to argue that "This could explain Attrill et al.'s (2008) results in the English Premier League as Liverpool, Manchester United, and Arsenal are historically some of the most successful teams and are all wearing red." (p. 16).

Specifically in soccer, Kocher and Sutter (2008) studied 306 games during the 2000/1 season of the top German division. Wearing red at home (or away) did not propel teams to winning more in comparison to the average league winning percentage in the home and away locales. In the Spanish Professional Football League (2001/02 to 2008/09 seasons), García-Rubio et al. (2011) also found that when endowments of resources (i.e., game attendance) and the ability of managers were accounted for, teams in red did not win more than other teams and surmised in regards to Attrill et al. (2008) that it was based on spurious correlation.

Allen and Jones (2014) relied on a more recent EPL game data (1992/93 to 2011/12, $N = 7720$), partially corresponding with Attrill et al. (2008) and the current investigation. They found that while red wearing teams had an improved average league position than non-red competitors, the red superiority effect failed to emerge in home winning percentage. The researchers were perplexed with how to reconcile the contrasting findings and reasoned that either (1) there was no effect altogether or that (2) "… changes in psychological functioning associated with viewing or wearing red stay with team members after shirt colour has been changed." (p. 13). We find the latter argument puzzling since it implies that the effect is not detected when the red uniform color is worn but emerges when the red uniform color is not worn.

Based on these studies it is clear that even when uniform color effects are detected, they tend to be inconsistent and small in magnitude (García-Rubio et al., 2011; Piatti et al., 2012). It is thus incumbent on future researchers in the field to study these effects while taking into account important team resource measures (e.g., market value of the team or budget, stadium capacity) to analyze the residual variances (the remaining variances) in order to assess if uniform colors effects indeed persist. Pointedly, Peeters and van Ours (2021) found that the relative home advantage was positively related to within-team variation in attendance. In line, in Study 1, Liverpool which dominated Everton in average league position and derby play played in Anfield which had a 53,000+ capacity while its cross-town rival played in Goodison Park with only 39,000+ seats. Manchester United, consistently ranked higher than Manchester city played in Old Trafford (~75,000 capacity) while its rival who moved to the Etihad in 2003 still had the much smaller stadium (55,000+).

The data and analyses presented in the current investigation are naturally not without their weaknesses. In the introduction we presented three major challenges to the ubiquity of the data analyzed in Attrill et al. (2008). While no major rule changes were instituted during the current period of study and all night games were now played in well-lit stadiums, the influx of foreign players occurred following the foundation of the EPL. The Bosman ruling in 1995, established free movement of labor including international soccer players (Szymanski, 2010) and currently only about half of all Premier players come from outside the UK. If color interpretation is culturally dependent (Sorokowski et al., 2014) this shift may be of importance in the interpretation of the results. However, Study 2, the first attempt to explore uniform color effects in a multi-country/cross-cultural context detected similarly null results in line with Study 1.

Another weakness which often hinders uniform colors research in team sports is the lack of consideration of what the away club is wearing. Attrill et al. (2008) never considered this in their analysis, nor did the present investigation, which followed in their footsteps. Indeed, teams are associated with their "traditional" uniform colors and the presence of home crowds may imbue certain colors with added "force" but uniform colors worn by the visiting team are certainly not of trivial matter as they are clearly perceived by all (i.e., players, referees, spectators). Recent basketball research, which controlled for the uniform colors of both teams in competition (Goldschmied & Lucena, 2018; Goldschmied & Spitznagel, 2020) still failed to detect any effects.

Original research discovering counter-intuitive effects tend to be exceedingly cited and Attrill et al. (2008) is no exception (183 citations on google scholar as of July 2021) as well as receive wide coverage by influential popular press outlets such as the BBC, CBS and the New York

Times (Jones, 2020; Schwartz, 2005; Stafford, 2012). This in contrast to the rather moderated exposure to the multitude of studies subsequently debunking the effect (61 citations of Allen & Jones, 2014, p. 29 citations of Kocher & Sutter, 2008, p. 52 citations of García-Rubio et al., 2011). This discrepancy may propel the lay-public to discount the critics and assume that the phenomenon is empirically well established. However, the current and most comprehensive research effort failed again to replicate the red uniform superiority effect in soccer. We conclude that given the accumulation of research in this sport, the effect detected by the original study is most likely based on specious association.

Credit author statement

Nadav Goldschmied: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing, Supervision and project management, Writing - Review & Editing and Visualization. **Philip Furley:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing, Supervision and project management, Writing - Review & Editing and Visualization. **Shakira Trejo**: Software, Data curation, Formal Analysis, and Investigation. **Angela Haddad**: Software, Data curation, Formal Analysis, and Investigation. **Aaron Böning**: Software, Data curation, Formal Analysis, and Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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